Questions:

- How much room is needed under the tables/how much room can the new legs occupy?
- Will the legs need physical replacement or just supports?
- How unstable are the legs? Do they get in the way of work and performance? If so, how much?
- How often/frequently will we be able to come in and work, will we be able to take measurements another time?
- How much weight does the table need to be able to hold (what is put on top of it)
- What are the heaviest things that are frequently on the table?
- Do you have the dimensions of the current table legs?
- Do you have the dimensions of the table?
- Does every leg on every table need support or changing?
- When/what would be the best way to contact you?
- Does appearance matter for the legs of the table? If so, what should they look like?
- How long have the current legs/table been there?
- What kind of wood is the current table made out of?
- Is there a budget/ how much money are you willing to spend?

Task 2:

Everyone is able to ask questions whenever

- ➔ Cooper: direct the conversation, introduce everyone and what we know so far and ask the primary questions
- ➔ Darian: keep track of time and taking photos of all the tables and their legs, note down anything that sticks out to him when looking at the table legs, initial ideas/solutions, potentially take measurements if we aren't able to another time
- ➔ Abel: record notes as well as update the document to new information, make sure to note communication methods and how often/available Mr.L is

Task 3: N/A

Task 4:

Question List: 5–7 vetted, open-ended questions, role confirmation, email stakeholders.

Good afternoon Mr.Lincicome,

We are a group of three engineering students from Mr Gregory's 6th period class. We would like to come in for 10 minutes to discuss and learn about the issues with your tables in the back. When is a good time to stop by to go over the problem with you? We have a number of questions about the tables and we look forward to your response and working with you.

Sincerely, Team CAD (Cooper G., Abel S, Darian S)

The main purpose of element A is to succinctly define the stakeholders problem. You do this by interviewing the stakeholder, analyzing the responses, restating their problem, and justifying why it is a problem in the first place.

The 6 W's: Who is involved, What are they doing, Where is it happening, When does it occur, Why does it happen, and how often? The who is the students in the classes he has, Mr. L, robotics students, kids that come into his room at any time, learn etc. The what is that they are using the table to do work, rest things on, stand on, relax on, and storage. The where is in the braces and the legs(everything except for the tabletop). When is a load or force onto the table from the top that requires the legs support. Why is because the braces and legs that are attached to the tabletop are weak, and in some places coming unassembled and falling apart leading to instability, wear/decreased durability, and potential safety and performance concerns when using the table. How is it has been happening for the past 5 years and it happens every once in a while, requiring Mr. L to tighten the screws on the braces.

The students of Mr. L's class are affected by the poor table, possibly making it hard for them to work. Mr. L is affected, as he is the beneficiary and will now be able to comfortably use the table. The custodians will have a safer time cleaning the STEM room if the table is fixed. The robotics club members would have a safer time using the table as well. Finally, those individuals who go into the classroom to interact with friends would also have a safer time in the case they sit on the table.

'If the problem isn't solved, Mr. L will need to find other replacements/tables which will cost a lot of money and effort classtime, decreased productivity due to concerns about the table, loss of storage due to people not being able to put things on the table tops"

Mr. L needs replacement legs and braces for the tables in the back, that must leave room under the table and can fully support the maximum load of the table without and give.

List 5 high priority design requirements

Requirements must be measurable. If you say "It must be strong," that will score a 1. If you say "It must support a 5lb vacuum hose without sagging," that'll score a 4-5.

It must support Cooper and Abel's full weight without giving, meaning the braces of the legs don't wobble/move

It must have enough clearance to fit a chair base fully under it (pushed in)

When applied with a repeated amount of student(s) pounds of force 5 amount of times to the table, the legs don't move more than 1cm

The design is replicable, can Mr.L read through the instructions/diagrams we provide him without any questions

It must be easily fixable. If screws are loosened/leg falls off, it must take less than 1 minute to put back on.

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